

# SARAH SEPEHRIAMIN

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## SUMMARY

MSc Biotechnology student at the University of Chester with a Bachelor's degree in Cellular and Molecular Biology. Has skills in both wet-lab techniques and computational biology. Brings experience in scientific research, publication, and interdisciplinary problem-solving. Highly motivated to apply scientific knowledge and innovative approaches in biotechnology, healthcare, and environmental applications.

## EDUCATION

- (Feb 2025- Feb 2027) MSc, Biotechnology with Professional Project/placement, University of Chester, UK
- (Sep 2018- Feb 2022) BS, Cellular and Molecular Biology, Azad University, Tehran

## PUBLICATIONS

(2024) Quercetin and its nanoformulations as promising agents for lung cancer treatment: A focus on molecular mechanisms

Drug Delivery Science and Technology, Elsevier

<https://doi.org/10.1016/j.jddst.2024.105933>

## LABORATORY SKILLS

- Molecular Biology Techniques:** DNA/RNA extraction, PCR/qPCR assay design and validation, electrophoresis and nucleic acid quantification, colony enumeration, restriction-enzyme digestion simulation, and cloning verification.
- Cell Biology Techniques:** Aseptic mammalian cell culture, cell viability assays, protein expression analysis, and cryopreservation.
- Biochemical Techniques:** ELISA, Western blot, spectrophotometric and fluorometric assays, chromatography (HPLC)
- Microbiology and QC:** Environmental monitoring, bioburden and endotoxin testing, media preparation, stability testing, and sterility assurance under ISO 17025 and GMP standards.
- Instrumentation and Analytical Tools:** Calibration and maintenance of pipettes, spectrophotometers, and balances; use of automated liquid handlers; data logging via ELN systems.
- Quality and Compliance Practices:** Adherence to GLP/GMP, ISO 13485, and MCERTS; preparation of SOPs, deviation reports, CAPA documentation, and audit-ready records ensuring full traceability and data integrity.

## TECHNICAL SKILLS

- Advanced Research Computing (ARC):** Experience using Linux-based ARC/HPC environments for large-scale bioinformatics analyses, workflow execution, software environment management (Conda), and reproducible genomic data processing.
- Version Control and Reproducible Research:** Git and GitHub for workflow organization, code management, and reproducible bioinformatics analyses.
- Artificial Intelligence:** Knowledge of Large Language Models (LLMs), prompt engineering, and generative AI applications in bioinformatics and computational biology.
- NGS Data Analysis and Bioinformatics Pipelines:** End-to-end analysis of high-throughput sequencing data (RNA-seq, scRNA-seq, ChIP-seq, WGS, WES), including quality control (FastQC, MultiQC), preprocessing (Trimmomatic, Cutadapt), alignment (HISAT2, STAR, Bowtie2, BWA), post-alignment processing (Samtools, BEDTools), variant calling (GATK, FreeBayes), and differential expression analysis (DESeq2, edgeR, limma).
- Workflow Automation and HPC Computing:** Development of scalable and reproducible bioinformatics pipelines in Linux/Unix environments using bash, awk, and sed. Proficient in workflow management systems (Snakemake, Nextflow) and containerization technologies (Docker, Singularity) with environment management via conda/Bioconda. Experienced in parallel computing, resource optimization, and large-scale data processing on HPC clusters.
- Genomic Data Analysis (GWAS and Variant Analysis):** Implementation of GWAS pipelines, including genotype QC and association testing using PLINK, population structure analysis (PCA), and visualization (Manhattan and QQ plots). Strong understanding of linkage disequilibrium, multiple testing correction, and genotype-phenotype association analysis.
- Single-Cell Transcriptomics:** Analysis of single-cell RNA-seq data using Seurat (R) and Scanpy (Python), including preprocessing, normalization, dimensionality reduction (PCA, UMAP), clustering, marker gene identification, and cell-type annotation. Experience with dataset integration and batch correction methods.
- Functional Genomics and Pathway Analysis:** Pathway enrichment and network analysis using Cytoscape and R/Python frameworks, including integration of multi-omics datasets and interpretation of biological pathways (e.g., KEGG) for

functional insights.

- **Statistical Analysis and Data Visualization:** Advanced data analysis in R (tidyverse, DESeq2, edgeR, ggplot2, pheatmap) and Python (pandas, numpy, matplotlib, seaborn) for statistical modeling, data transformation, and high-quality visualization of multi-omics data.
- **Reproducibility and Data Standards:** Implementation of reproducible research practices, including pipeline validation, version control, documentation, and adherence to FAIR data principles.

## PROJECTS AND EXPERIENCE

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- **Global Prevalence of Hepatitis B Virus Resistance Mutations**  
University of Chester — Supervised by Dr Navaneethan Palanisamy (2026)  
Developing a bioinformatics pipeline for large-scale analysis of HBV genomic sequences to identify clinically relevant resistance mutations. Processed GenBank/FASTA datasets using Python (Biopython), including sequence filtering, genotyping, alignment, and translation. Performed mutation screening and frequency analysis, and visualized genotype-specific mutation patterns using R (ggplot2, dplyr) to support downstream epidemiological insights.
- **scRNA-seq Analysis of Craniosynostosis**  
University of Oxford — Supervised by Dr Stephen Twigg — 2026-present  
Conducted single-cell RNA sequencing (scRNA-seq) analysis to investigate cellular heterogeneity and transcriptional changes associated with craniosynostosis. Processed and analyzed scRNA-seq datasets using Seurat/R, including quality control, normalization, clustering, dimensionality reduction, and cell-type annotation.
- **scRNA-seq Analysis of Human Diabetic Kidney Disease**  
Conducted scRNA-seq analysis of DKD kidney datasets using Seurat/R, including QC filtering, normalization, clustering, UMAP visualization, marker identification, pathway enrichment, and cell-cell communication analyses.
- **Whole Exome Sequencing Analysis of Drug-Resistant TNBC**  
Processed raw WES data from SRA (SRR) through QC, alignment, and BAM processing using Samtools and standard pipelines. Performed variant calling with GATK (HaplotypeCaller/Mutect2), followed by filtering, annotation (ANNOVAR/SnpEff), and prioritization. Conducted pathway enrichment analysis (e.g., KEGG) and visualized variants using IGV and Cytoscape to identify mutations associated with drug resistance.
- **RNA-seq Transcriptomic Analysis**  
Performed end-to-end RNA-seq analysis including quality control, alignment, and differential expression analysis using DESeq2/edgeR. Interpreted transcriptional changes in the context of biological pathways and disease mechanisms.
- **TCGA Data Retrieval and Analysis**  
Retrieved and processed RNA-seq datasets using TCGAbiolinks. Performed normalization, batch correction, and statistical analysis for cohort comparisons, and generated visualizations including heatmaps and expression profiles.
- **Molecular Docking for Drug Target Interaction Analysis**  
Applied molecular docking approaches using AutoDock Vina to investigate binding interactions between candidate compounds and disease-related protein targets. Analyzed binding energies and interaction networks, and visualized docking results using PyMOL to support structure-based insights into drug response.

## TEACHING AND MENTORING EXPERIENCE

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- **Bioinformatician**  
University of Chester-(2025-Present)  
Working on computational biology and bioinformatics projects involving analysis of high-throughput sequencing data. Performed data preprocessing, quality control, and downstream analysis using standard bioinformatics tools. Utilized Linux environments, R, and Python for data handling, analysis, and visualization. Contributed to ongoing research projects in genomics and related fields.
- **Quality Assurance Analyst | Data Process Automation**  
Medley Pharma LTD, UK, 2025-Present  
Contributed to QA activities within a pharmaceutical environment, while developing automated data extraction and documentation workflows using Python and Document AI to improve efficiency, reduce manual data entry, and ensure data integrity and compliance with GMP standards.
- **Clinical Research Assistant**  
Naft Hospital, Tehran (2023–2024)  
Supported clinical research projects, including patient recruitment, data collection, and management of clinical datasets. Assisted with experimental design and analysis of molecular and genomic data. Ensured compliance with ethical guidelines and Good Clinical Practice (GCP) standards. Contributed to study reports, data visualization, and publication preparation.
- **Research Assistant**  
Azad University, Tehran, 2019–2022  
Assisted in molecular biology and bioinformatics research projects, performing DNA/RNA extraction, PCR/qPCR, and gel electrophoresis. Supported data analysis using R and Python for transcriptomic datasets. Prepared reports, maintained accurate laboratory records, and contributed to publication-quality results.
- **Bootcamp in Laboratory Techniques**  
Biograd Company, Liverpool, 2025

Completed intensive training in molecular biology techniques including DNA/RNA extraction, PCR/qPCR, plasmid preparation, and mammalian cell culture. Gained practical experience in sterile technique, cell maintenance, and experimental design, with an emphasis on accurate documentation and reproducibility.

## AWARDS

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- Recognised by faculty for exceptional technical precision and analytical accuracy in MSc Biotechnology projects.
- The Excellent Semester Accomplishment Award in the "Department of Cellular and Molecular" (B.Sc. GPA: 3.679 out of 4.000) at Islamic Azad University.

## ACADEMIC CONFERENCES AND SEMINARS

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- **(2025) Postgraduate Research Conference – Department of Natural Sciences, University of Chester**  
Participated in academic discussions and research presentations showcasing postgraduate work in the field of natural sciences.
- **(2025) Bioreactor and Growth Environment Workshop – Keele University**  
Participated in a hands-on workshop focused on bioreactor systems, microbial cultivation, and optimization of growth environments for industrial and research applications.
- **(2025) Participant, TechSkill Company Event**  
Attended a seminar on "The Importance of Programming Languages in Today's World", focusing on the role of coding skills in modern technology and industry.

## INTERESTS AND OTHER INFORMATION

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As a graduate student with a solid foundation in Cellular and Molecular Biology, I am passionate about understanding the complexities of biological systems and exploring various areas within the life sciences. My research interests span Computational Biology, Data Analysis, Quality Control, Environmental Sustainability, Biotechnology, Biochemistry, Pharmaceutical Science, Cell and Molecular Biology, Genetics, and Genomics. I am adaptable to diverse environments and thrive when actively engaged in meaningful work alongside my education. I am highly responsible, detail-oriented, and committed to delivering high-quality results in all my endeavors. <https://www.linkedin.com/in/sarah-sepehri-amin/>